



TalentRank AI

Explainable AI Candidate Discovery Engine — turning a job description and a candidate pool into a transparent, evidence-backed shortlist.

Data & AI Challenge

Intelligent Candidate Discovery

INDIA RUNS Hackathon

PROBLEM

Recruiters can't review candidate pools deeply enough

Traditional applicant-tracking workflows lean on keyword filters. That approach misses semantically relevant candidates, over-ranks shallow keyword matches, and gives recruiters little basis to defend a shortlist decision.

Keyword filters are brittle

A candidate who says "led a team" instead of "management experience" gets skipped, even with the right background.

Rankings aren't explainable

Recruiters need evidence — matched skills, experience fit, gaps — not a black-box score.

Sensitive data creeps into scoring

Name, gender, age, and similar fields must never influence a ranking decision.

A recruiter decision-support workflow, not a black box

1. JD Intelligence

Rule-based extraction of role, seniority, must-have / nice-to-have skills, responsibilities, domain, and experience range.

2. Privacy-safe profiles

Candidate and job text built only from approved ranking fields — sensitive attributes are never added.

3. Hybrid retrieval

TF-IDF + Sentence-Transformer (MiniLM)
semantic similarity + skill coverage narrow the pool to a top-50 shortlist.

4. Optional CrossEncoder rerank

Pairwise relevance scoring over the top-50 for higher-fidelity ordering when latency allows.

5. Explanation cards

Every ranked candidate gets a fit summary, matched/missing skills, evidence, and a risk statement.

6. Responsible AI report

Every run writes an audit trail of used features, excluded fields, and the decision-support disclaimer.

Design choices, and the reasoning behind them

Hybrid over pure semantic

Semantic-only retrieval misses exact skill/tool matches recruiters expect to see (e.g. "AWS"). TF-IDF keeps lexical precision; embeddings add meaning. Combining both outperforms either alone on a small, noisy candidate pool.

Retrieve-then-rerank, not rerank-everything

CrossEncoder scoring is accurate but slow on CPU. Restricting it to the top-50 hybrid shortlist keeps latency practical while still getting the most accurate model where it matters most.

Explainability as a first-class output

A ranked list without justification isn't useful to a recruiter who has to defend a shortlist decision — so every candidate carries structured evidence, not just a number.

Fairness enforced in code, not policy

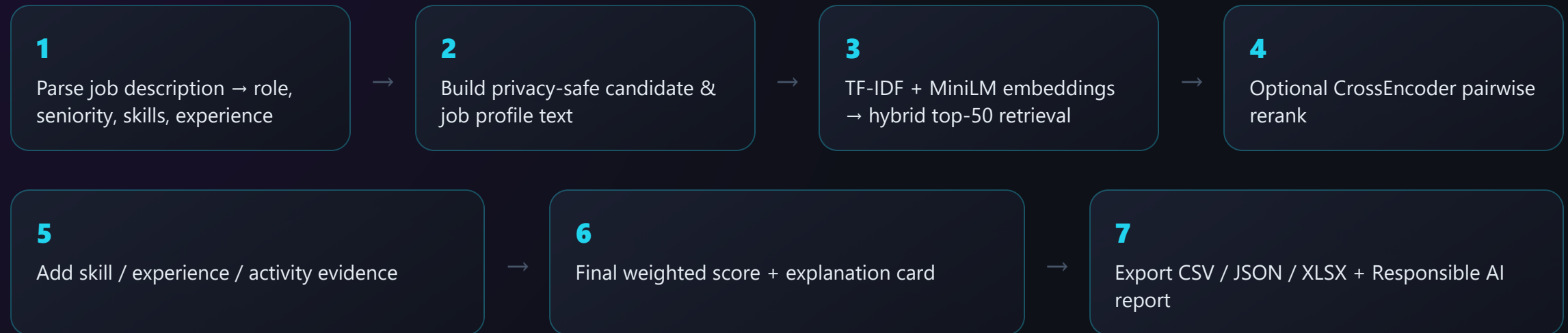
`src/fairness.py` excludes name, gender, religion, caste, age, DOB, photo, marital status, and address from profile text and scoring — structurally, not by convention.

Streamlit for the demo surface

Fast to build, easy for judges to run locally, and good enough for a recruiter-facing decision-support tool at hackathon scope.

HOW IT WORKS

End-to-end pipeline



Scoring formula (CrossEncoder enabled)

```
final_score = 0.30 × cross_encoder_score + 0.20 × skill_match_score + 0.20 × semantic_score  
             + 0.15 × tfidf_score + 0.10 × experience_match_score + 0.05 × activity_score
```

When CrossEncoder is disabled, its 30% weight redistributes equally to semantic similarity and skill coverage.

TECH STACK & EVALUATION

Built with a standard, auditable ML stack

Python 3.11

pandas / NumPy

scikit-learn TF-IDF

Sentence-Transformers MiniLM-L6-v2

CrossEncoder ms-marco-MiniLM-L-6-v2

Streamlit

pytest

Evaluation metrics reported by the pipeline

P@5 / P@10

Precision at top-K

Recall@10

with relevance labels

NDCG@10

ranked relevance quality

MRR

first-relevant-hit rank

Without labels: average top-10 skill coverage, semantic score, experience match, and ranking latency — plus an ablation comparing TF-IDF only, Semantic only, Hybrid, and Hybrid + CrossEncoder.

Home & Ranked Candidates

 **TalentRank AI**
Explainable AI Candidate Discovery Engine

Demo pages

Home

JD Intelligence

Ranked Candidates

Candidate Explanation

Evaluation

Responsible AI

Export Output

Active job

J001 — Data Scientist

Shortlist size

5

☐ Use Cross-Encoder Reranking

Run ranking

 **TalentRank AI**
Explainable AI Candidate Discovery Engine

🌟 Explainable candidate discovery for fast, defensible shortlists

Combine job intelligence, semantic relevance, skills, experience, activity, and optional CrossEncoder evidence in one recruiter-ready workflow.

This system is a recruiter decision-support tool. Final hiring decisions should be human-reviewed.

Active role

Data Scientist

Available candidates

10

Jobs

Must-have skills

Seniority

Mode

4

5

Mid

Hybrid

▼ Demo flow

1. Explore JD Intelligence

2. Run ranking

3. Inspect Ranked Candidates

4. Open Candidate Explanation

5. Review Evaluation, Responsible AI, and Export Output

 **TalentRank AI**
Explainable AI Candidate Discovery Engine

Demo pages

Home

JD Intelligence

Ranked Candidates

Candidate Explanation

Evaluation

Responsible AI

Export Output

Active job

J001 — Data Scientist

Shortlist size

5

☐ Use Cross-Encoder Reranking

Run ranking

 **Ranked Candidates**

Shortlist ordered by explainable evidence

Shortlist

5

Top score

83.5

Average score

59.0

Latency

0.24s

Podium

#1

C001

Strong fit

83.5

final score

#2

C002

Moderate fit

61.1

final score

#3

C009

Moderate fit

60.3

final score

Full shortlist

rank	candidate_id	final_score	cross_encoder_score	semantic_score	tfidf_score	skill_match_score	experience_match_score	activity_score
1	C001	83.45	None	76.7	48.06	100	100	88
2	C002	61.09	None	69.37	13.43	60	100	76
3	C009	60.27	None	64.51	20.7	80	33.33	65
4	C004	45.41	None	52.89	17.58	40	66.67	72
5	C005	44.99	None	65.15	11.25	20	100	70

7 / 9

Candidate Explanation & Score Breakdown

TalentRank AI

Explainable AI Candidate Discovery Engine

Demo pages

Home

JD Intelligence

Ranked Candidates

Candidate Explanation

Evaluation

Responsible AI

Export Output

Active job

J001 — Data Scientist

Shortlist size

5

Use Cross-Encoder Reranking

Run ranking

Candidate Explanation

Explanation card and score breakdown

Select candidate

#1 — C001

Rank

#1

Final score

83.5

Confidence

High

Strong fit

Candidate C001 is a strong fit because they match 5/5 required skills and have 4 years of experience for the 3-6 year role range. Projects are listed, but no direct job-skill evidence was detected.

Matched must-have skills

Python, SQL, pandas, scikit-learn, Machine Learning

Matched nice-to-have skills

streamlit, AWS

Missing required skills

None

Confidence level

High

Evidence and risk details

Experience: Has 4 years of experience for the 3-6 year role range.

Projects: Projects are listed, but no direct job-skill evidence was detected.

Activity: Activity score of 88.0/100 indicates strong engagement.

Category	Score
Final	83.5
Semantic	76.7
Keyword	48.1
Skills	100.0
Experience	100.0
Activity	88.0
CrossEncoder	0.0

8 / 9

Honest about scope, clear on the roadmap

Current limitations

- Sample data is small and not representative of production hiring volume
- Skill extraction is rule-based and depends on taxonomy coverage
- Scores are decision-support signals, not job-performance predictions
- CrossEncoder reranking adds noticeable CPU latency

Future improvements

- Expand the skill ontology with role/industry-specific aliases
- Multilingual resume and JD support
- Recruiter feedback loop for calibrated relevance labels
- Vector-store retrieval for larger candidate pools
- Auth, audit trails, and RBAC for production deployment

This system is a recruiter decision-support tool. Final hiring decisions should always be human-reviewed.